

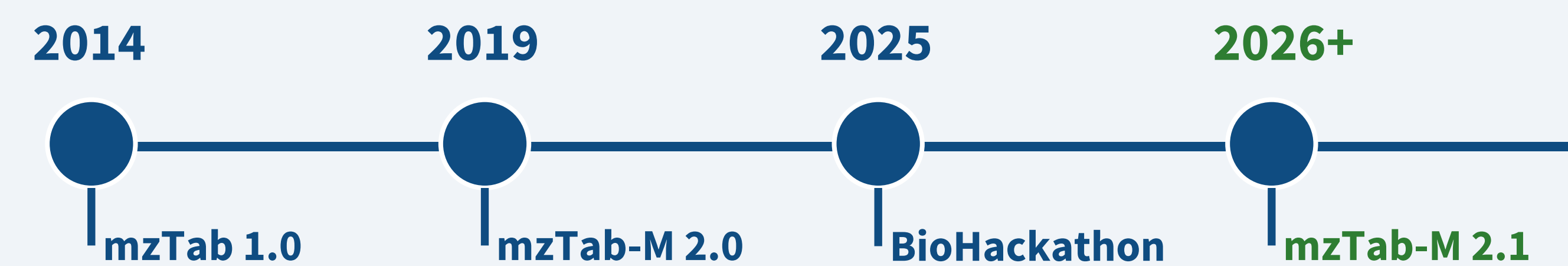
Towards mzTab-M 2.1 – Evolving the HUPO-PSI standard format and ecosystem for reporting of small molecule mass spectrometry results

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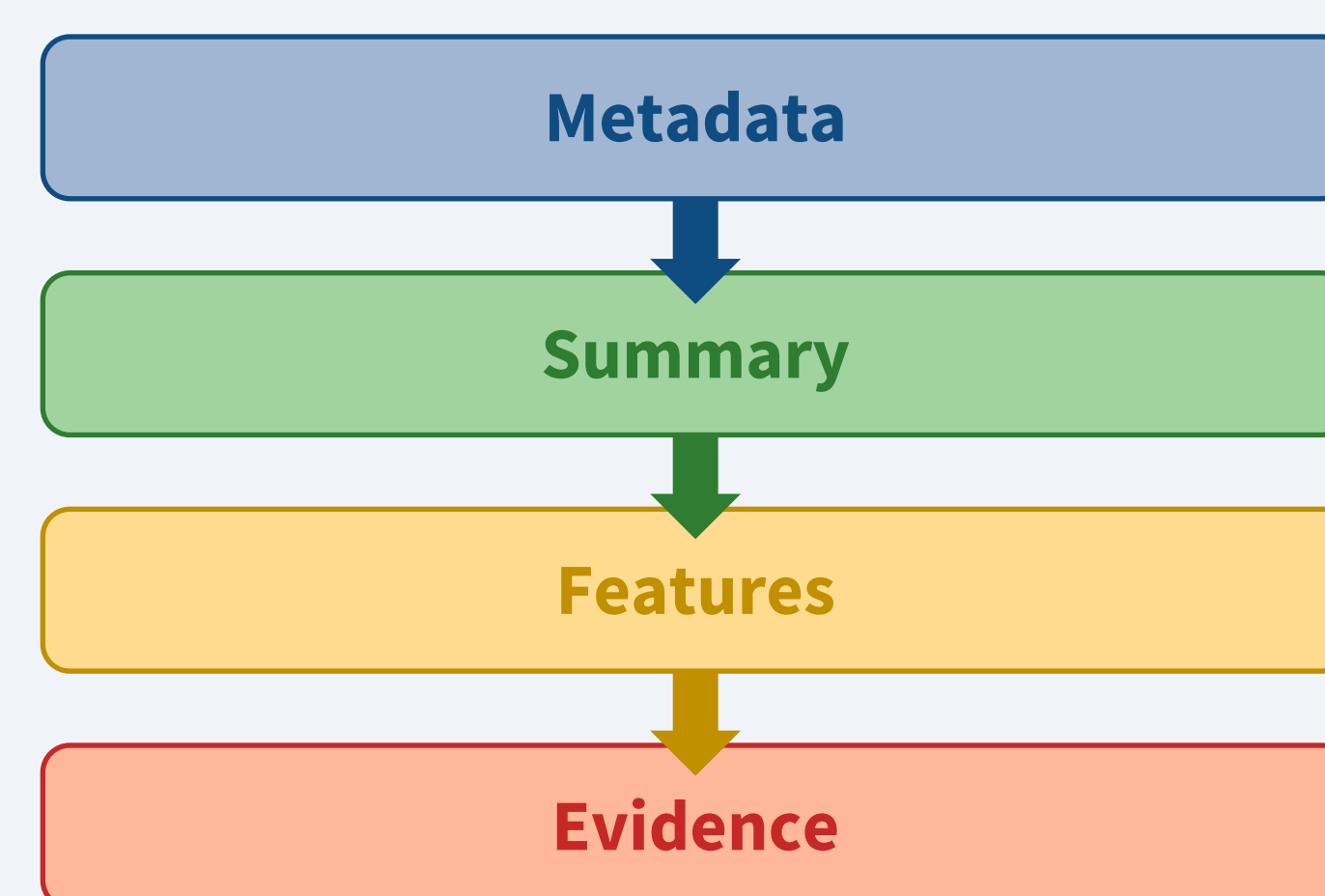
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INTRODUCTION

- The **mzTab format family** was developed by the Proteomics Standards Initiative (PSI) to capture the design and both qualitative and quantitative results of mass spectrometry experiments
- mzTab 1.0** for Proteomics: adopted in some tools and as submission format for EMBL-EBI PRIDE repository (2014)¹
- mzTab-M 2.0**: new implementation for Metabolomics tools and data repositories (published 2019)²
- Uses **ontologies** for metadata and experimental outcomes (processed results) and for semantic validation
- Interoperability between all producers and consumers was **not systematically evaluated**
- Limited community and vendor uptake due to limited repository support**
- Through different **joint activities**, we are **updating mzTab-M to version 2.1** and at the same time **reaching out to software developers** (academic and commercial) to **support implementation** of mzTab-M



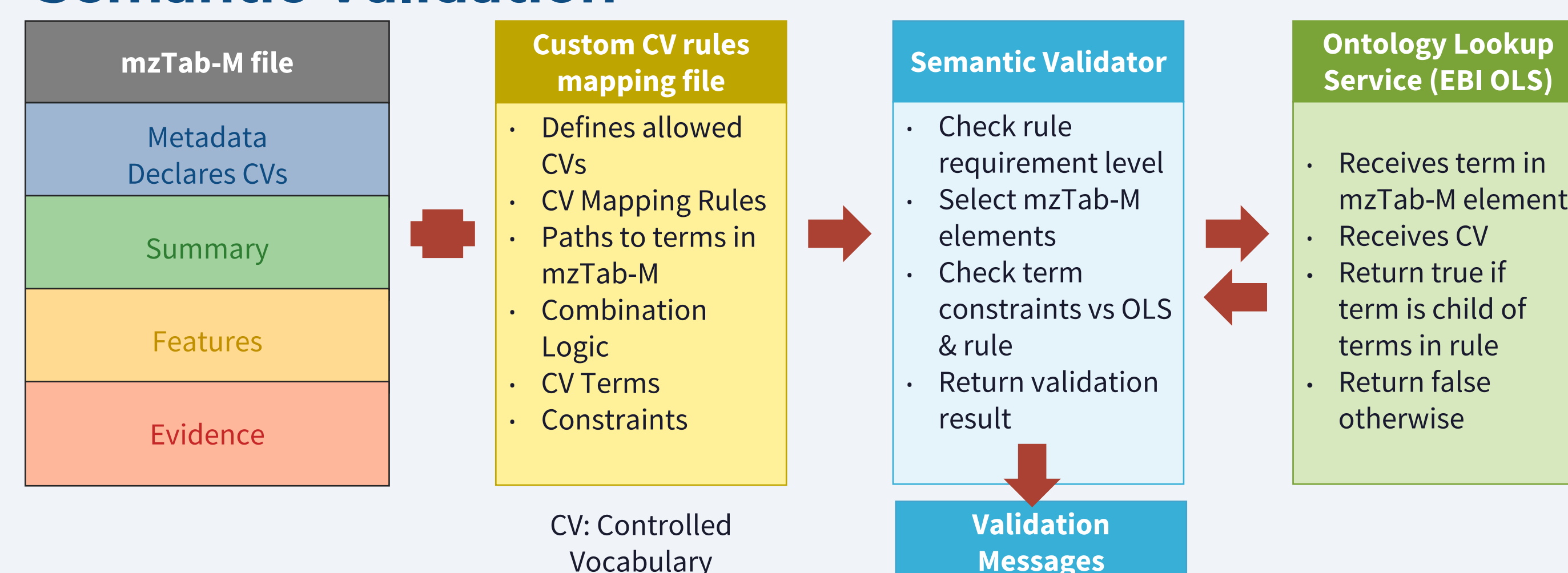
FORMAT STRUCTURE & VALIDATION



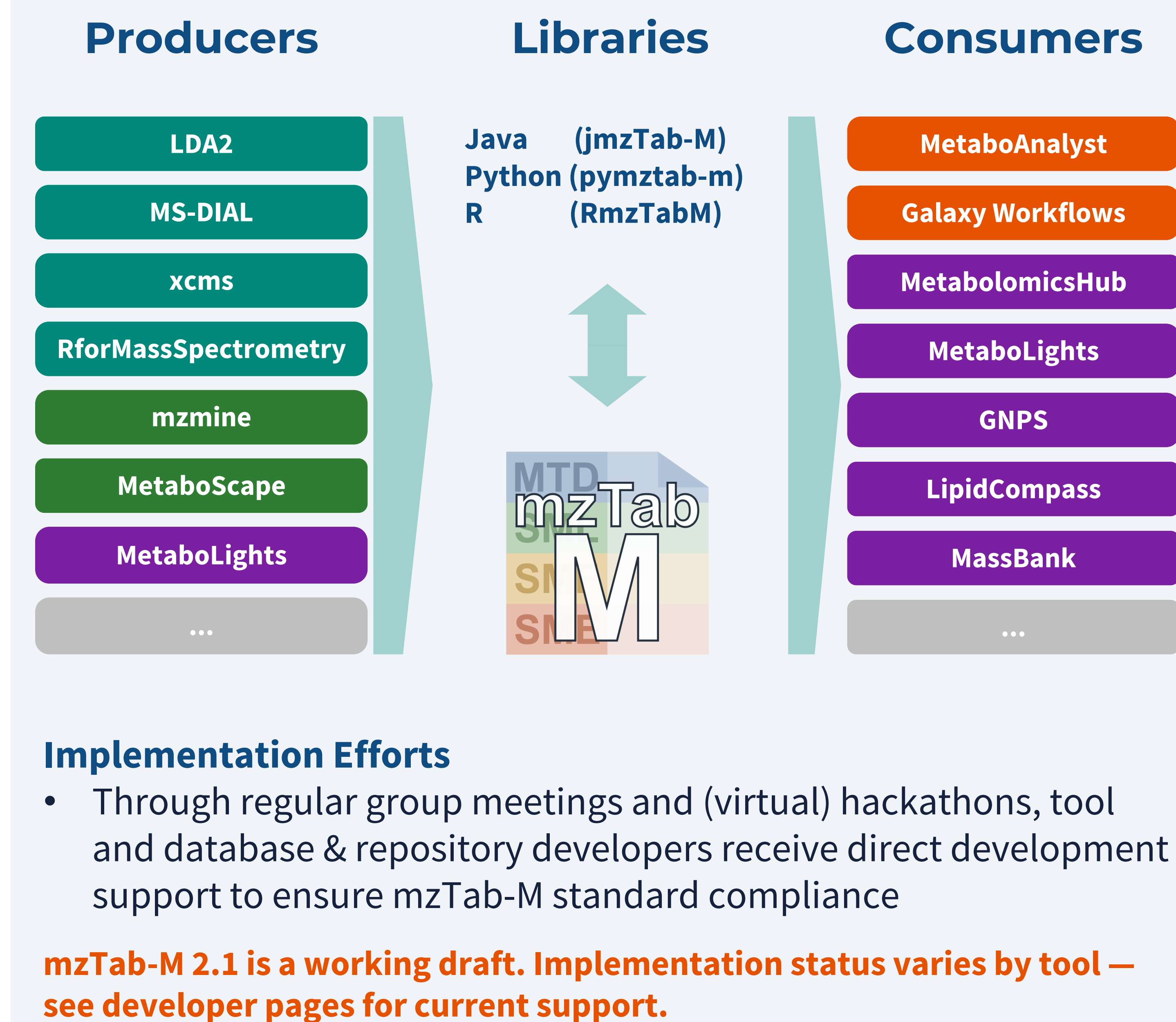
Features

- schema-backed, (de)serialization from/into JSON or TSV formats
- CV-controlled metadata, contacts, samples, assays, MS-runs, study variables, quantification
- MS features, e.g. m/z, RT, CCS
- Identification evidence, e.g. DB hits, similarities, additional info, reference MS scans in run (mzML)

Semantic Validation



mzTab-M 2.1 SOFTWARE ECOSYSTEM ROADMAP



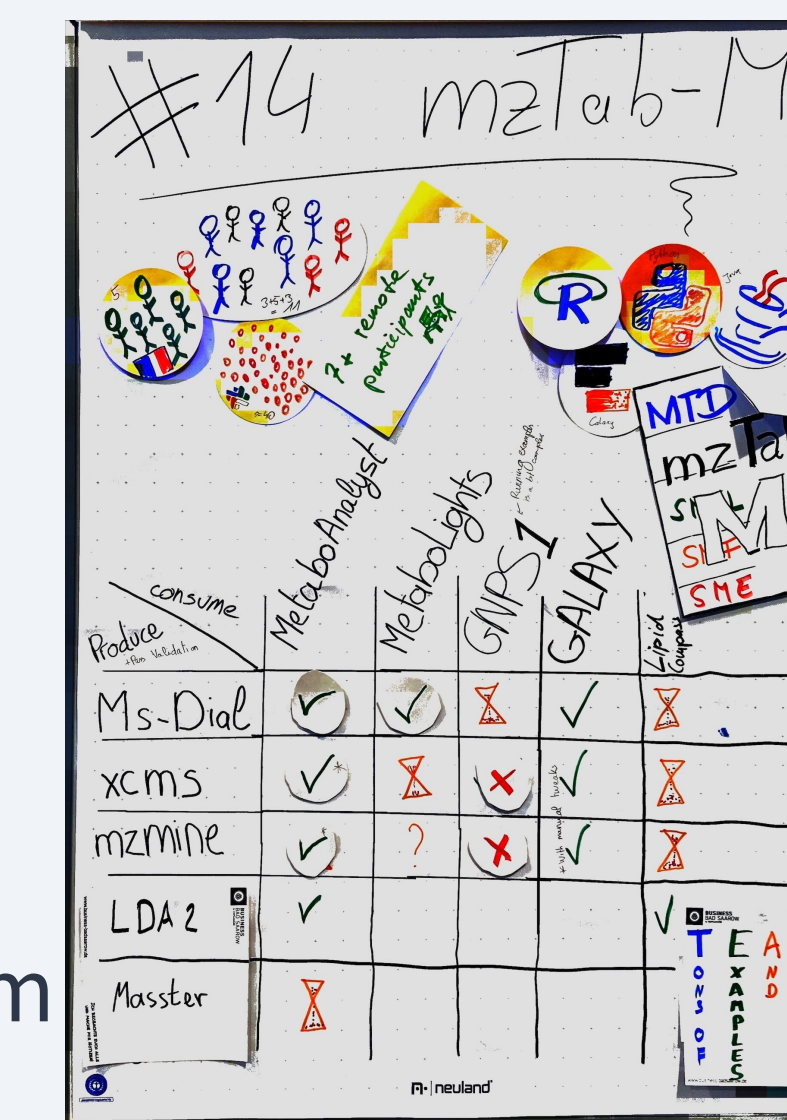
mzTab-M 2.1 DRAFT

- The mzTab-M 2.1 draft remains largely backward-compatible with 2.0 while adding new features and resolving several issues.
- Machine-Readable & Agent-Friendly**
 - ✓ Semantic enrichment through use of CV parameters
 - ✓ Agentic-AI skills to create mzTab-M files
 - Study Variable Groups**
 - ✓ Added study variable groups for complex experimental designs
 - Interoperability with other HUPO-PSI Standards**
 - ✓ Scan reference on evidence level to mzMLs³
 - ✓ Integration of Universal Spectrum Identifier (**USI**) for mass spectra⁴
 - ✓ Referencing of **mzSpecLib** mass spectral libraries and peak annotation format⁵
 - ✓ Referencing of **mzQC** for mass spectral quality control data⁶
 - Protocol Sections**
 - ✓ Sample processing protocols
 - ✓ Chromatography details
 - ✓ Mass spectrometry acquisition parameters
 - Improved Validation**
 - ✓ Enhanced semantic validation profiles and interoperability tests
 - Improved Documentation**
 - ✓ Updated website structure for users and developers
 - ✓ New examples and updated links to implementations

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INTEROPERABILITY EXERCISE – KEY RESULTS

- ✓ Established interoperability between multiple producers and consumers of mzTab-M
- ✓ Collected broad test collection of mzTab-M files from diverse sources
- ✓ Tested compatibility across tool combinations
- ✓ Reported errors and contributed fixes upstream



Issues Identified

- Examples missing or not sufficient to demonstrate complex setups
- Inconsistencies in how tools represent study variables and metadata, need to enable iterative refinement of files in workflows
- Missing protocol information for sample processing, chromatography, and MS details
- Integration with other tools and resources

REFERENCES & ACKNOWLEDGMENTS

References

- Griss *et al.* (2014) MCP, 10.1074/mcp.O113.036681 (mzTab 1.0)
- Hoffmann *et al.* (2019) Anal. Chem., 10.1021/acs.analchem.8b04310 (mzTab-M 2.0)
- Martens *et al.* (2010) MCP, 10.1074/mcp.R110.000133 (mzML)
- Deutsch *et al.* (2021) Nat. Methods., 10.1038/s41592-021-01184-6 (USI)
- Klein *et al.* (2024) Anal. Chem., 10.1021/acs.analchem.4c04091 (mzSpecLib & mzPAF)
- Bielow *et al.* (2024) JASMS, 10.1021/jasms.4c00174 (mzQC)

Acknowledgments

We would like to thank the following people for their help, support and feedback: Luca De Martino, Fiona Hui, Jeff Xia, Eisuke Hayakawa, Mingxun Wang, Pieter Dorrestein, Robin Schmid, Tomas Pluskal, Helge Hecht, Denise Sclenter, Nikolas Kessler, Sagar Gore, Anna Eggers, Norman Storz, Guillaume Patoine, Johan Gustafsson, Michał Burdukiewicz. This work was carried out as part of the HUPO-PSI Metabolomics working group activities. It was further supported as project 14 “Proteomics and Metabolomics File Format Interoperability Fest” at the BioHackathon Europe 2025, Berlin, funded by ELIXIR Europe. This work was supported by the German Network for Bioinformatics Infrastructure (de.NBI) and ELIXIR-DE (Forschungszentrum Jülich and W-de.NBI-006). The authors declare no competing financial interest.

Resources

- HUPO-PSI mzTab-M specification: github.com/HUPO-PSI/mzTab-M
- Documentation Hub: hupo-psi.github.io/mzTab-M
- MetaboLights: www.ebi.ac.uk/metabolights
- MetabolomicsHub: www.metabolomicsshub.org/

Related Posters

TP 319, WP 447, WP 520

